MYKHAILO BARYBIN

GAMEPLAY PROGRAMMING

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Computer Science student focused on gameplay systems in **C++ and C#**. Project work includes **deformable terrain**, custom co-op networking, enemy navigation and runtime graphics tools.

EDUCATION

Sheridan College | Computer Science, Game Engineering
Bachelor's degree in progress | Sep 2023 - Expected 2028

SELECTED TECHNICAL PROJECTS

How to Dig 2026

Unity, C#, TCP/UDP | Personal project in development

- Developing a 1-4 player co-op excavation game with loot recovery, equipment, earnings quotas and persistent progression.
- Terrain deformation uses a density field, marching tetrahedra and asynchronous chunk mesh generation to rebuild excavated geometry.
- Custom host-authoritative TCP sessions and UDP LAN discovery synchronize gameplay; height-aware goblin navigation supports ladder and rope traversal.

SDLTagYouAreIt 2026

C++, SDL3 | Gameplay coursework | github.com/Lethache/SDLTagYouAreIt

- Normalized input and deltaTime-based movement drive player control; NPC seek/flee distance bands connect to animation states and collision-triggered tagging.
- Object-pool reuse manages entity lifecycles; binary save/load preserves player and NPC state.

DirectX 12 Lighting and Tools 2026

C++, HLSL, C++/CLI, WinForms | Graphics coursework

- Interactive Phong lighting, camera/light controls and position-based colour modes; a tools DLL exposes runtime controls to the C++ application.

SDLFileChunks 2026

C++, SDL3 | Systems coursework | github.com/Lethache/SDLFileChunks

- Command-pattern image-buffer operations with undo/redo stacks and binary reconstruction of TGA image data from file chunks.

TECHNICAL SKILLS

Languages C++, C#, HLSL, Python, JavaScript
Development Unity, SDL3, DirectX 12, MonoGame, WinForms, Unreal Engine
Workflow Git/GitHub, Visual Studio, debugging, regression testing, Blender

EXPERIENCE

Trucker Budget | QA / Marketing Specialist, Contract | Apr 2026 - Present
Test iOS functionality, investigate data and paywall defects, document reproduction steps and verify fixes.

Aventra Solutions | Project Coordinator / Quality Lead | Feb 2025 - Present
Coordinate requirements and delivery; review web and 3D work and track issues through resolution.